Axolotl 0.3.0 on Linux

Installing Axolotl and working with the axl command line compiler

1. What you get

Axolotl is a JVM language that translates to readable Java 21. Its compiler checks that values which may be null are handled, using NullAway, and moves every error back to the line of the `.axl` file.

The installer does three things:

- installs the `axl` command together with its own Java 21 runtime, so `axl` needs no Java on the computer. The runtime has every module of Java SE, AWT and Swing included;
- puts `axl` on the `PATH`;
- runs `axl setup`, which copies the Axolotl libraries and the Gradle and Maven plugins into the local Maven repository, so that Gradle and Maven projects find them.

2. Requirements

- A 64-bit x86 Linux with glibc.
- For the `.deb`: Debian 12 or Ubuntu 22.04 or newer, or a system based on them, and `sudo`.
- For the tarball: any such Linux; no root is needed.
- Gradle and Maven, if you use them, run on a JDK 21 of their own: the runtime inside `axl` is only for `axl`.

3. Installation

Debian and Ubuntu: the .deb package

Download `axolotl-0.3.0-linux-amd64.deb` from https://github.com/succubus-uwu/axolotl/releases/tag/v0.3.0 and install it with `apt`. The `./` matters: without it `apt` looks for a package of that name in its repositories.

```
$ cd ~/Downloads
$ sudo apt install ./axolotl-0.3.0-linux-amd64.deb
```

The package installs into `/opt/axolotl`, registers `/usr/bin/axl` with `update-alternatives`, pulls in the X11, ALSA and font libraries the Java runtime uses, and runs `axl setup` for the user who called `sudo`.

Any Linux: the tarball

Download `axolotl-0.3.0-linux-x64.tar.gz`, unpack it and run `install.sh`. It needs no root and installs into `~/.local/share/axolotl`, with `axl` linked from `~/.local/bin`:

```
$ tar xzf axolotl-0.3.0-linux-x64.tar.gz
$ sh axolotl-0.3.0/install.sh
```

`AXL_HOME` chooses another directory: `AXL_HOME=~/tools/axolotl sh axolotl-0.3.0/install.sh`. If `~/.local/bin` is not on your `PATH` yet, the script says so; add this line to `~/.profile` and log in again:

```
export PATH="$HOME/.local/bin:$PATH"
```

> Programs that open windows or play sound through Java need the X11 and ALSA libraries of your system; the `.deb` installs them, the tarball leaves that to you.

4. Checking the installation

```
$ axl --version
axl 0.3.0
```

`axl --help` lists the commands, and `axl help <command>` or `axl <command> --help` describes one.

5. The first program

Sources live in a source directory, with one directory per package part, as in Java. Make a project directory with `src/app/Main.axl` in it:

```
package app

import java.util.List

fn greet(name: String?) -> String = "Hello, " + (name?.trim() ?: "stranger")

fn main() {
    val words = List.of(args)
    println(greet(if (words.isEmpty()) null else words.get(0)))
}
```

Run it from the project directory. Everything after the file name goes to the program as `args`:

```
$ axl run src/app/Main.axl
Hello, stranger
$ axl run src/app/Main.axl Ann
Hello, Ann
```

`axl run` finds the source directory from the package of the file, translates and compiles every `.axl` and `.java` file in it, checks nulls and runs `fn main()`. Save the files in UTF-8.

6. The commands

Command	What it does
`axl run <file> [args...]`	Compiles the source tree of the file and runs its `fn main()`.
`axl check <dirs...>`	Translates and compiles, keeps no output and prints the errors. Fast feedback, and what editors call.
`axl build -d <dir> <src>`	Translates and compiles a source directory into class files.
`axl retrace [file]`	Rewrites the `.java` lines of an old stack trace to `.axl` lines.
`axl setup`	Installs the libraries and plugins into the local Maven repository.
`axl help [command]`	Describes the commands.

Options

Option	Commands	Meaning
`-cp`, `--classpath`	run, check, build, retrace	Jars and class directories the code uses, separated by `:`.
`-d <dir>`	build	Where the class files go.
`-s <dir>`	build	Where the generated `.java` and `.axlmap` files go (default: the `-d` directory).
`--strict-null`	run, check, build	Treats values from Java code without nullness annotations as nullable.
`--no-null-checks`	run, check, build	Skips Error Prone and NullAway: faster, but nulls are not checked.
`--inline-elvis`	run, check, build	Expands `?:` at statement level without a lambda.
`--no-param-checks`	run, check, build	Leaves out the null checks of public functions and constructors against callers from Java.
`--no-smap`	build	Keeps the line numbers of the generated `.java` files in the classes.
`-m`, `--map-dir`	retrace	A directory with generated `.java` and `.axlmap` files; repeatable.
`--repository`	setup	The local Maven repository to install into.
`-h`, `-V`	all	Help and version.

7. Checking code: axl check

Add a file that forgets a null check, `src/app/Broken.axl`:

```
package app

fn shout(name: String?) -> String = name.toUpperCase()
```

```
$ axl check src
src/app/Broken.axl:3:42: error: value `name` may be null: use `?.`, `?:` or check `if (name
    != null)` [AXL102]
```

Every message has the form `file:line:column: error|warning: text [code]`. The codes are listed in `docs/errors.md`: AXL0xx come from Axolotl's own checks, AXL1xx from javac, Error Prone and NullAway, reworded in Axolotl terms. Warnings do not stop a build.

`axl check`, `axl build` and `axl run` exit with code 1 when there are errors and 0 otherwise, so scripts and CI can test `$?`. `axl check` takes several source directories, such as `axl check src/main src/test`.

Fix it with `name?.toUpperCase() ?: ""` and `axl check src` prints nothing.

8. Building class files: axl build

```
$ axl build -d out/classes -s out/gen src
```

`out/classes` gets the class files, `out/gen` the generated Java and the `.axlmap` line maps. The generated Java is plain Java 21 you can read. Code that uses other libraries passes them with `-cp`, separated by `:`.

The classes run on any Java 21 or newer with the Axolotl runtime library on the classpath. `axl setup` put it into the local Maven repository. `axl` does not provide a `java` command, so this needs a Java of your own:

```
$ RT=~/.m2/repository/org/axl/axl-runtime/0.3.0/axl-runtime-0.3.0.jar
$ java -cp "out/classes:$RT" app.Main Bob
Hello, Bob
```

For anything larger than a few files a Gradle or Maven build is simpler (section 10).

9. Stack traces

Classes built by `axl build` name the `.axl` file and line themselves, so a program started with `java` prints them in every stack trace and log:

```
Exception in thread "main" java.lang.IndexOutOfBoundsException: Index: 5 Size: 1
    ...
    at app.Main.first(Main.axl:5)
    at app.Main.main(Main.axl:8)
```

`axl retrace` is for traces that still name the generated Java, such as old logs or classes built with `--no-smap`. It reads a trace from a file or from standard input and points it at the `.axl` lines using the `.axlmap` files (`-m out/gen`):

```
    at app.Main.first(Main.java:9)
    at app.Main.main(Main.java:12)
```

```
$ axl retrace -m out/gen trace.txt
    at app.Main.first(Main.axl:5)
    at app.Main.main(Main.axl:8)
```

10. Gradle and Maven projects

After `axl setup` both build tools find Axolotl in the local Maven repository. They still download Error Prone, NullAway and the other dependencies from Maven Central, and need a JDK 21.

Gradle

```
// settings.gradle.kts
pluginManagement { repositories { mavenLocal(); gradlePluginPortal() } }
dependencyResolutionManagement { repositories { mavenLocal(); mavenCentral() } }

// build.gradle.kts
plugins {
    application
    id("org.axl.lang") version "0.3.0"
}
application { mainClass = "app.Main" }
```

Sources go to `src/main/axl` and `src/test/axl`; `gradle run` builds and runs the program.

Maven

A new project from the archetype, ready to build with `mvn package`:

```
$ mvn archetype:generate \
    -DarchetypeGroupId=org.axl -DarchetypeArtifactId=axl-archetype-quickstart \
    -DarchetypeVersion=0.3.0 \
    -DgroupId=org.example -DartifactId=demo -Dpackage=org.example.demo
```

11. Updating and uninstalling

Install the new `.deb` with `sudo apt install ./...deb`, or run the `install.sh` of the new tarball, which replaces the earlier installation.

- The `.deb`: `sudo apt remove axolotl`.

- The tarball: delete `~/.local/share/axolotl` (or your `AXL_HOME`) and `~/.local/bin/axl`.

The libraries `axl setup` installed stay in the local Maven repository under `org/axl`; delete that directory too if no project needs them.

12. Troubleshooting

Problem	What to do
E: Unable to locate package from `apt`.	Write the path with `./` (or a full path): `sudo apt install ./axolotl-...deb`.
`apt` cannot satisfy the dependencies on an older release.	Use the tarball, which depends on nothing.
axl: command not found after the tarball.	Add `~/.local/bin` to the `PATH` (section 3) and open a new terminal.
Maven or Gradle cannot find `org.axl.lang` or `axl-maven-plugin`.	Run `axl setup` as the user who builds; check that the build lists `mavenLocal()`. If you set `-Dmaven.repo.local` in `MAVEN_OPTS`, run `axl setup` again from that shell, or pass `axl setup --repository <dir>`.
`java` is not found, or is older than 21, when running built classes or Gradle.	Install a JDK 21; `axl` keeps its own runtime to itself.
A compile takes long the first time.	`axl` starts a JVM and loads javac and NullAway on every call; a few seconds are normal. `--no-null-checks` skips NullAway for quick experiments.

More

The documentation of the language, the Gradle and Maven plugins and the editors is at https://github.com/succubus-uwu/axolotl/tree/v0.3.0/docs. Releases and installers: https://github.com/succubus-uwu/axolotl/releases.